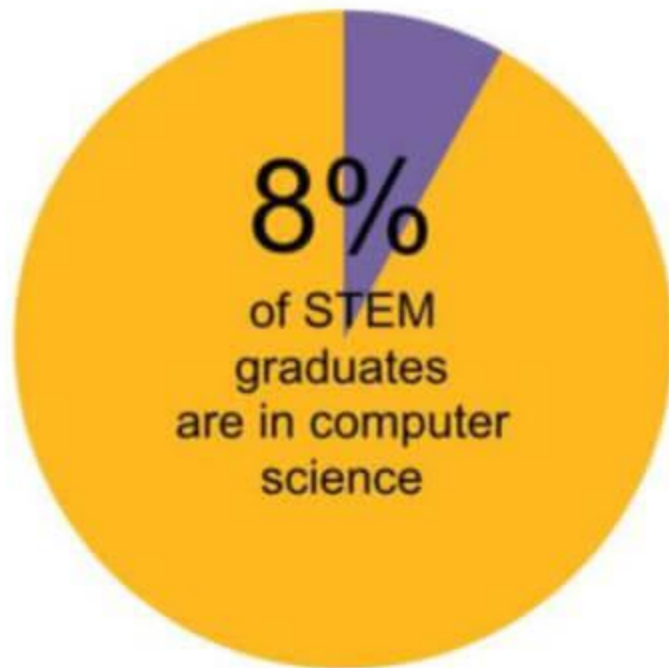
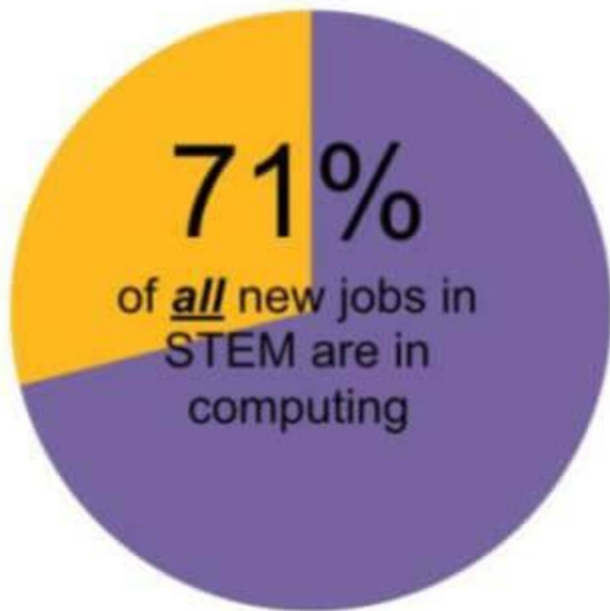


Careers in Computer Science

CSCI 1101



The STEM Opportunity is in Computer Science



Computer Science Graduates Earn 40% More



Many Emerging Fields Are Driven by Software



Artificial
Intelligence



Data
Science



Self Driving
Cars



Drones



Computational
Biology



Analytics



Robotics



Video
Games



VFX
Hollywood



Digital
Art

Software Engineer/Developer

What is the difference?



Design, develop, test, and maintain software applications.

Average Salary: \$90,000 - \$140,000+

High demand across various industries including tech, finance, healthcare, and more.

Computer Languages: C++, Java, Python

Software Engineer, YouTube



YouTube



Bengaluru, Karnataka, India



Early

Minimum qualifications

- Bachelor's degree or equivalent practical experience.
- 1 year of experience with software development in one or more programming languages (e.g., Python, C, C++, Java, JavaScript).

[Learn more](#)

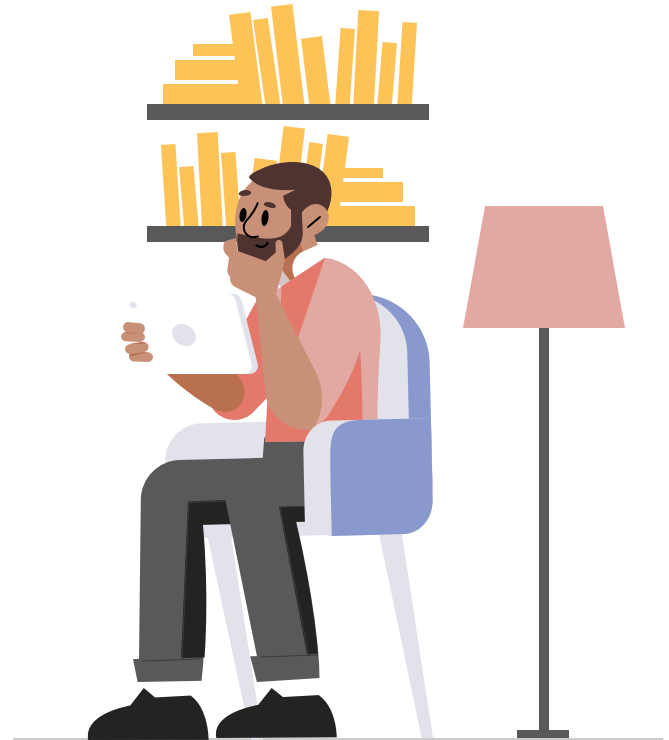
Machine Learning Engineer

Develop algorithms and models that allow machines to learn and make decisions.

Average Salary: \$110,000 - \$170,000+

Rising, especially with advancements in AI and automation.

Computer Languages & Frameworks: Python, Pytorch, TensorFlow, Git



Staff Software Engineer, Machine Learning/Gen AI, Chrome



Google



Seattle, WA, USA; Kirkland, WA, USA; +1 more



Advanced

Minimum qualifications

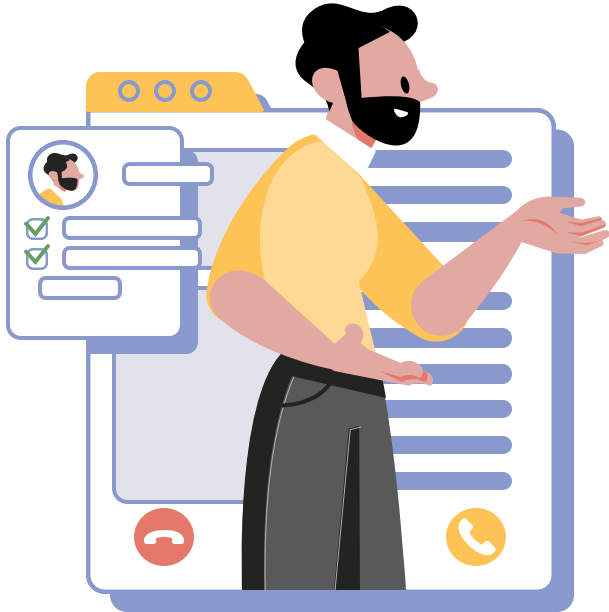
- Bachelor's degree or equivalent practical experience.
- 8 years of experience in software development, and with data structures/algorithms.
- 5 years of experience with machine learning algorithms and tools (e.g., TensorFlow), or applied ML (e.g., deep learning, natural language processing).
- 5 years of experience testing, and launching software products, and 3 years of experience with software design and architecture.
- Experience in Python programming.

[Learn more](#)

Roadmap

Skill	Est. Time	Learning Phase
Programming (Python)	1-2 months	Beginner
Version Control (Git)	1-2 weeks	Beginner
Data Structures & Algorithms	1-2 months	Beginner
SQL	1-2 months	Beginner
Mathematics and Statistics	2-3 months	Beginner
Data Handling and Processing	1-2 months	Intermediate
Machine Learning Fundamentals	3-4 months	Intermediate
Advanced Machine Learning Concepts	2-3 months	Advanced

Cybersecurity Analyst/Engineer



Protect organizations from cyber threats by implementing security measures.

Average Salary: \$90,000 - \$150,000+

Critical as cybersecurity becomes increasingly vital across industries.

Computer Languages and Frameworks: C/C++, Python, Java, JavaScript, SQL, Bash/Shell Scripting

Other Skills: Networks

Security Engineer, Vulnerability Detection



Google



Warsaw, Poland



Early

Minimum qualifications

- Bachelor's degree in a technical field (e.g., Cyber Security, Computer Science) or equivalent practical experience.
- Experience in C/C++ source code security analysis.
- Experience coding in one or more coding languages (e.g., Java, Python, Go).
- Experience working in device, OS, or embedded system security domain.

[Learn more](#)

Full-Stack Developer



Work on both the front-end (user interface) and back-end (server-side) of web applications.

Average Salary: \$80,000 - \$140,000+

High demand, particularly in startups and smaller companies.

Computer Languages: C++, JavaScript, Python, Node.js, Go, React, Objective-C, Ruby, SQL, PHP

Full Stack Developer

Tube Force · Montreal, QC (Hybrid)

 Easy Apply

Save



Requirements:

- Proven expertise in Laravel, Node.js, Python, Go, Etc...
- You have strong Front End skills and are fluent in developing in React.js, Next.js etc..
- Deep understanding of video processing and encoding/transcoding.
- Ability to progressively integrate React.js/Next.js environments into existing projects.
- Perfectly comfortable with a "you build it, you own it" mindset and you understand your responsibilities continue after you push code to production
- Value and seek individual contributions, but you also understand that team deliverables have a more significant and stronger impact

If you're ready to be part of a team that's shaping the future of web development and video processing, we want to hear from you!

 Apply now and let's create the future together! 

DevOps Engineer

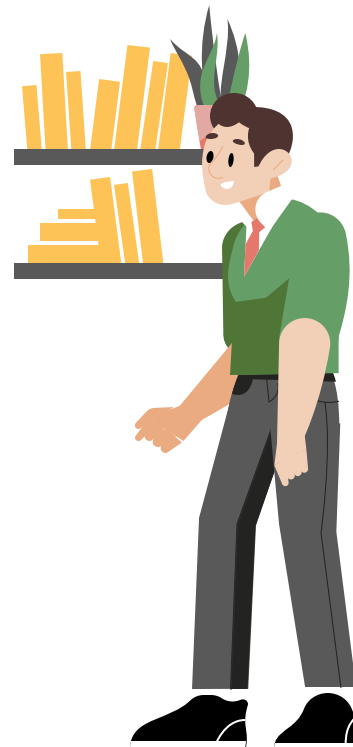
Streamline software development and IT operations through automation and collaboration tools

Average Salary: \$95,000 - \$150,000+

High demand due to the need for efficient and continuous software delivery.

Computer Languages & Frameworks: Python, Bash/Shell Scripting, Ruby, Go (Golang), JavaScript

Other Skills: Git, Continuous Integration/Continuous Deployment (CI/CD), Docker, AWS, Networks



DevOps Engineer

VXG · Canada (Remote)

 Easy Apply

Save

...

WHAT YOU NEED TO HAVE

- Knowledge of git on admin level
- Understanding of software development cycle, principles of CI/CD, experience with Gitlab CI
- Experience in administrating Linux systems
- Knowledge of Docker and Kubernetes
- Knowledge of bash or Python
- Understanding of computer network technologies and network protocols (IP, TCP, UDP, TLS)

WHAT WILL BE GOOD TO HAVE

- Knowledge of software development tools: GNU make, meson, autotools, scons, gradle;
- Experience with AWS: EC2, AutoScaling, CloudWatch, RDS, S3, CloudFormation;
- Experience with infrastructure monitoring tools (Prometheus, Zabbix);
- Administrating logging services (ELK);
- Experience with configuration management systems (Ansible, Chef) and infrastructure management systems (Terraform);
- Experience with VMWare vSphere.

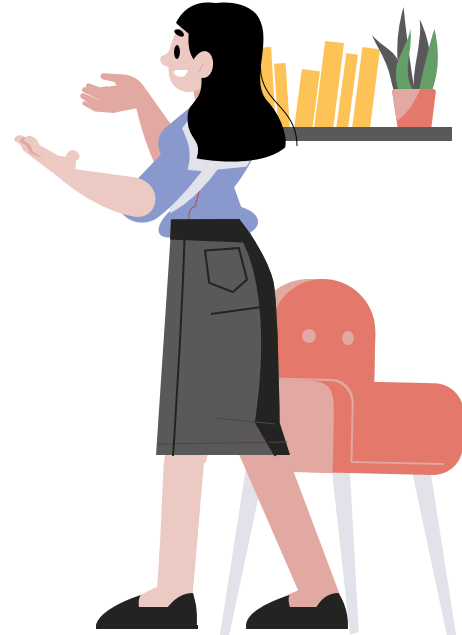
Database administrator

Manage and maintain databases, ensuring their performance, security, and availability.

Average Salary: \$80,000 - \$130,000+

High demand, especially in industries that rely heavily on data storage and management.

Computer Languages & Frameworks: Bash, Python, Java, PHP, Oracle, SQL Server, DB2, Pearl



Junior Database Administrator

Leidos · Alexandria, VA (Hybrid)

Apply 

Save



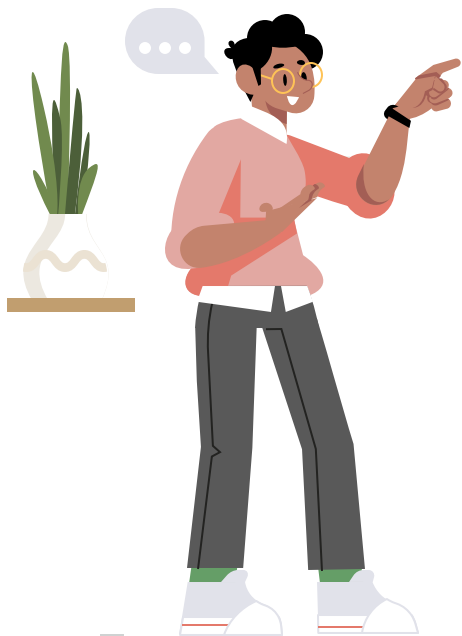
Basic Qualifications

- Education: Bachelor's Degree and 2 years of experience OR Master's Degree and 0 years of experience. 4 years of additional work experience, OR Associate's Degree plus 2 years of experience, may be substituted for a Bachelor's Degree.
- Degree in computer science, Information systems, Engineering, business or related scientific or technical discipline.
- Excellent collaboration, communication, and interpersonal skills.
- An active Secret level security clearance with the ability to obtain a TS/SCI clearance is required for this role.

Preferred Qualifications

- Database certifications (i.e. Oracle; SQL server etc.) or other related technical certifications.
- Experience with scripting languages such as Bash, Python, Ruby, or PHP and Windows Power Shell or Pearl.
- Familiarity with GIS, geospatial data, and geospatial products, including Esri.

Virtual Reality (VR) / Augmented Reality (AR) Developer



Design and develop immersive virtual and augmented reality experiences.

Average Salary: \$90,000 - \$180,000+

Growing rapidly with the expansion of VR/AR applications in gaming, education, healthcare, and more.

Computer Languages and frameworks: C/C++, C#, HTML, JavaScript, Unity, OpenGL

Software Developer, Augmented Reality

Google · Mountain View, CA (On-site)

Apply 

Save



USA.Minimum qualifications:

- Bachelor's degree or equivalent practical experience.
- 2 years of experience programming in C++, C, and C#.
- Experience developing APIs and integrating third-party services.
- Experience developing with the Unity game engine.

Preferred qualifications:

- Master's degree or PhD in Computer Science or related technical field or equivalent practical experience.
- Experience with AR, virtual reality(VR), XR, and mixed reality(MR) Platforms.
- Experience with open graphics standards such as OpenGL or Vulkan.
- Proficiency in Math with respect to graphics.

Game Developer/Engineer

Game developers design, develop, and test video games across various platforms, including consoles, PCs, and mobile devices. They work on game mechanics, graphics, AI, and user interfaces

Average Salary: \$75,000 - \$120,000+

High demand, particularly in large gaming studios, indie game development, and emerging platforms like VR/AR.

Computer Languages & Frameworks: C++, C#, and JavaScript, game engines: Unity, Unreal Engine, and Godot

Other Skills: understanding of 3D modeling, animation, and physics; strong problem-solving skills and creativity.



Rocket Science Group 5.0★



Software Engineer - Unreal

Albany, NY

\$80K - \$130K (Employer est.)

⚡ Easy Apply

What We Value:

Strong Technical Skills: Proficiency in C++ and significant experience with Unreal Engine, including a deep understanding of game architecture, optimization, and 3D math.

Engineering Excellence: A commitment to writing clean, maintainable, and efficient code, with a focus on quality and best practices.

Feature Ownership: Experience as the primary owner of a feature or system within a shipped product.

Team Collaboration: Excellent communication skills and the ability to work collaboratively with cross-functional teams.

Problem-Solving: Strong analytical and problem-solving skills, with a proactive approach to identifying and addressing technical challenges.

Passion for Gaming: A genuine passion for video games and a desire to contribute to creating engaging and enjoyable experiences for players.

Interest in Learning: Eagerness to learn about other technologies and practices to stay current and versatile.

What You May Also Have:

Console Experience: Experience developing for current-generation consoles (e.g., Xbox Series X/S, PlayStation 5) is a plus.

Mentorship Skills: Previous experience coaching, teaching or mentoring less experienced engineers to learn new technical skills.

Communication Skills: Previous experience working with internal and external stakeholders to gather feedback, provide support, and problem solve.

Other Careers in Computer Scientists



Information Security Analysts

Computer and Information Research Scientists



Web Developers

Digital Designers



UX/UI Engineers



Computer Systems Analysts

Computer Network Architects



Business Analysis

And more ...

Major Employers of Computer Scientists

Microsoft

Microsoft hires tech professionals in roles like software engineer, data scientist, and solution architect.

Alphabet

Alphabet is the parent company of Google. Google employs various computer science professionals, including software engineers, UX researchers, and software developers.

Apple

Apple offers computer science jobs in hardware, software and services, and machine learning and AI.

US DoD

DoD jobs for people with computer science degrees include cyberthreat analysts, machine learning scientists, and artificial intelligence engineers.

Amazon

Computer science professionals can find jobs in software development, software engineering, and data science at Amazon.

Meta

Meta hires computer science professionals for roles like computer research scientist, security software engineer, product designer, and data scientist.

Projected percent employment change, 2021–31

